

Week 04. Cause and Effect in the RCT. Hypothesis testing. T-tests. Exmples

Part 2: Statistics and Econometrics. MK68-2023/24

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- Ceteris paribus
- Hypothesis testing
- Examples of Hypothesis Testing
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- Effects of a de-worming pill
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Ceteris paribus

- Watch again: [Ceteris Paribus: Public vs. Private University](#)

Ceteris paribus I: RCT (1)

RCT - Randomized Control Trial (watch [Randomized Trials: The Ideal Weapon!!!](#))

- The treatment is assigned randomly in a controlled environment where all other things remain fixed.

Treatment group

- a random sample from a population, where a treatment is assigned.

Control group

- random sample from a population, without any treatment.

The Law of large numbers says:

- Both, the treatment and control groups are on average the same as they are sampled from the same population.
- The only difference between the two samples is the fact of treatment.

Causal effect = Average Causal Effect = Difference in averages between two groups

Ceteris paribus I: RCT (2)

Experiments are often **not available, costly, sometimes unethical**, and often illegal.

Think, about how to make experiments to estimate the effects of:

- years of schooling on earning?
- additional child on mother's employment?
- army service on the job perspective?
- electronics (phones/laptops) in class on students performance?
- health insurance on health

Ceteris paribus II: Regression

When experiments are not possible, **econometric methods** come to our rescue.

Regression

- **regressions** control the effects of other factors in the **causal effect of interest**.
- **regression** contains **multiple variables**, which **represent measurable and approximate not measurable** factors.
- Regression analysis has to be theoretically sound by ensuring that **key assumptions are validated**.

Ceteris paribus III: Innovation

Use clones!

Add treatment!



Econometrics measures the effect of a cause

Read more:

1. Cause and Effect, a teaser:
 - Ch 1.4 in ([Wooldridge 2020](#)).
2. The world of potential outcomes:
 - Ch 1 in ([Joshua D. Angrist and Pischke 2014](#)).
3. It is all about the research design:
 - ([Joshua D. Angrist and Pischke 2010](#)).
4. The RCT is not the ultimate solution in development economics:
 - ([Deaton 2009](#)).

Example (exam): think of an experiment (1)

Can training and education of parents about nutrition reduce children's stunting?

 What RCT experiment can help us answer this question?

Let us think of and discuss:

- An outcome;
- A treatment;
- The population;
- A sample;
- How to measure the impact of the treatment?

Example (exam): think of an experiment (2)

1. Outcome variable:
 - Stunting score of a child.
2. Treatment variable:
 - Parents' special training in nutrition.
3. Targeted population
4. Samples:
 - Randomly select eligible subjects.
5. Randomly assign treatment (communities with training) and control (community without it) groups
6. Collect the pre-intervention data;
7. Apply the treatment
8. Collect the post-intervention data;
9. Measure the effect

Question 1.

Does the **research design** imply that (if found) education has a causal impact on stunting?

- Answer:
 - **Yes!**
 - All other things are equal because the research design is based on the RCT.
 - **Training** is the only change between the treatment and control.

Question 2.

Is the stunting on average less in the treatment compared to the control?

- Answer:
 - The absolute number, yes. **BUT!**
 - Treatment and control are two different samples from the population.
 - Means vary naturally for random samples.
 - We **must apply statistical inference** about the population based on the estimates.
 - Conclusion: **We do not have sufficient information!**

Question 3.

Can we recommend education as one of the ways to reduce stunting?

- Answer:
 - If any effect is found, yes! But...
 - Results are only applicable to the population under investigation.

Conclusion

Stunting score		
Statistics	Control	Treatment
mean	-1.41	-1.33
sd	1.42	1.57

To conclude that

- two means estimated from different samples are different (same),
- we need to perform **statistical inference!**
- **Without statistical inference, we cannot know what are the chances (odds) that the difference in means is caused by a random variation in samples.**

Exam questions examples

- What is the difference between counterfactual and potential outcomes?
- What is the counterfactual? How is it different from the factual?
- What do we usually observe in the data, factual or counterfactual?
- What are the factual and counterfactual in the Urea fertilizers and Rice yields example?
 - How to generate a counterfactual?
- What kind of experiment can help answer the questions below? Think about such an experiment and discuss it.
 - Does the health insurance make people more healthy?
 - What is the causal effect of fertilizers on yields?
 - Does going to a private school improve earning?
- What makes randomized control trials such a powerful tool for revealing the causal effects?
 - What statistical principles ensure the ceteris paribus under the RCT?
- Give examples of measurable and not measurable factors that need to remain fixed in order to reveal the causal effect of fertilizer application on crop yields.

Hypothesis testing

A statistical hypothesis test is a **method of statistical inference** used to decide **whether the data** at hand **sufficiently supports a particular hypothesis**.

Hypothesis testing allows us to make probabilistic statements about population parameters.

- Read more in Appendix C-6 in ([Wooldridge 2020](#)).

Hypothesis testing

What?

- Hypothesis Test: A statistical test of the **null hypothesis** against an **alternative hypothesis** ([Wooldridge 2020](#)).

Why?

- We never know the true parameters of the population.
- We must infer about the population from samples.
- HT establishes certainty behind our inference (probability or an error).

How?

- Following Neyman-Pearson's ([Neyman and Pearson 1933](#)) approach to hypothesis testing.

Neyman-Pearson's approach

- assumes that we can perform an infinite number of repeated sampling based on population;
- aims to **minimize errors of making a false conclusion about the relationship**
- controls the probability of making an error, when we say “**we reject**” the **Null hypothesis**;

Hypothesis Testing: Steps

1. H_0 : Null/Zero Hypothesis;
 - $H_0 : \mu = 0$
2. H_1 : Alternative Hypothesis;
 - $H_1 : \mu \neq 0$
3. **Estimate parameters:**
 - $\hat{\mu} = \widehat{\text{mean}}$, and
 - $s = \widehat{SD}$
4. **Compute the test statistics;**
5. **Conclude:**
 - to reject $H_0 = \text{accept } H_1$, or
 - fail to reject $H_0 = \text{accept } H_0$
6. **Calculate probabilities of making an error.**
 - probability value (aka “p-value”)

Hypothesis Testing may result in two errors

Type I error

- false positive
- probability value: “p-value”

The error of rejecting H_0 , when H_0 is the **TRUE** hypothesis

Type II error

- false negative

The error of accepting H_0 , when H_0 is the **FALSE** hypothesis

⚠ NEVER KNOWN to the researchers!

p-value probability value

Neyman-Pearson's approach to HT suggests:

- reject H_0 , when
- p-value (probability of the **Type I error**)
- is below **the level of significance α** .

That means:

- when we reject H_0 , the probability that we've made a mistake is below α

Thus, **we reject H_0 only and only if p-value $< \alpha$** .

α is the level of significance

- Maximum probability of a Type I error in hypothesis testing.
- In social sciences, the level of significance should not be higher than 5% ($\alpha = 0.05$).

Example (exam): HT about stunting (1)

Can training and education of parents about nutrition reduce children's stunting?

Our data measures change in stunting levels between pre- and post-training for 500 individuals.

Answer the following questions:

- Formulate H_0 and H_1 .
- Provide examples of the Type I and Type II errors in the context of your hypothesis.

Solutions

Null hypothesis: -

$$H_0 : \Delta_{\text{stunting}} = 0$$

Alternative

hypothesis: -

$$H_1 : \Delta_{\text{stunting}} \neq 0 -$$

$$H_1 : \Delta_{\text{stunting}} > 0 -$$

$$H_1 : \Delta_{\text{stunting}} < 0$$

- Type I error:
 - We conclude **to reject** $H_0: \Delta_{\text{stunting}} \neq 0$,
 - But this observation is caused **by a random variation in the sample**.
 - In the population, training had nothing to do with stunting.
 - Repeated sampling will probably fail to show a difference.
- Type II error:
 - We conclude **to accept** H_0 : stunting rate did not change after training.
 - Training helped but, e.g., the food shortage deepened the malnutrition problem in the survey year.

Example (2): HT about ravens' color

Question: What % of ravens are White?

Hypothesis:

- H_0 : White ravens make up 0% of the population
- H_1 : Share of white raven is not zero ($> 0\%$)

Data:

- we went to a park;
- we saw 100 ravens (or crows);
- **no white ravens!**

Results: the share of white ravens is 0!

Conclusion: **fail to reject H_0 (Accept H_0)**

Example (2): Raven color

Conclusion: Accept H_0 (there are no white ravens)

Did we make any errors? If yes, which one?

- We've made the **type II error: false negative**.

Because:

- We do not know if H_0 is a true hypothesis.
- We did not use theory (did not study ornithology).
- **The theory has to justify our choice of the H_0 .**



Homework on hypothesis testing

For the research questions below:

1. Describe how would you measure the outcome variable.
 2. Formulate H_0 and H_1 .
 3. Discuss Type I and Type II errors in the context of your hypothesis.
 4. Provide examples of the factors, which could cause these errors to appear.
 5. Think of an RCT experiment that could help answer these questions.
- How students' performance in a class is different between years?
 - Are the wheat yields higher in the farms that use precision agriculture?

Research questions:

- What is the difference between the demographic characteristics of individuals in the control and treatment groups?
 - The characteristics are: age, weight, height, income, number of children, household size...

Exam questions examples (2)

- Formulate H_0 and H_1 .
- Provide an example of a type I / II error relevant to the hypothesis.
- What are the type I/II errors?
- Which error type is not known to the researcher that operates observational data? Why?
- When do we reject the null hypothesis? What kind of error we are trying to minimize?
- What is the significance level?

Examples of Hypothesis Testing

The paper ([Kuchenbecker et al. 2017](#)) aimed to assess the potential of community-based nutrition education to improve height-for-age z-scores in children 6–23 months of age.

- The authors performed the RCT experiment in Malawi over 5 years.
- Data was collected for the control and treatment groups in the baseline and the end-line.

Kuchenbecker et al. (2017)

- If the paper finds a difference between nutrition statuses in the control and treatment groups, does it mean that education has a causal effect on nutrition?
 - Yes, if treatment and control groups are the same on average in other factors.
 - Such factors are Income, Education level, HH size, Farming abilities, ...

Checks and balances

- How to identify if the factors are NOT DIFFERENT between the two groups?
 - Perform statistical hypothesis testing.
 - Formulate the hypothesis:
 - H_0 : a factor is the same across the two groups.
 - H_1 : a factor changes between two groups.
 - Fail to reject H_0 , when there is no difference

Checks and balances table

This is an essential table that evaluates if the treatment and control groups of the RCT are the same on average.

The table shows:

- estimated mean and SD in the control and treatment groups;
- statistical test about the difference in means;
- probability value of the type I error;

Kuchenbecker et al. (2017) Table 2

	Baseline 2011 (n = 832)					Endline 2014 (n = 959)				
	Control		Intervention		p^+	Control		Intervention		p^+
	n	mean ($\pm$ SD)	n	mean ($\pm$ SD)		n	mean ($\pm$ SD)	n	mean ($\pm$ SD)	
Household size	412	5.8 (2.3)	419	5.7 (2.1)	0.49	466	5.3 (1.9)	493	5.3 (2.0)	0.78
Wealth status	413	-0.2 (3.4)	418	0.5 (3.7)	0.01	466	-0.3 (3.8)	493	0.1 (3.7)	0.10
Years of education of caregiver	412	5.9 (3.0)	418	6.2 (2.9)	0.12	466	6.5 (2.9)	493	6.8 (2.7)	0.09
Years of education of household head	409	8.1 (2.5)	416	8.1 (2.4)	0.79	449	8.1 (2.7)	480	8.3 (2.7)	0.41

Also available [here](#).

- Formulate H0 and H1 about each variable.
- Perform statistical HT about the mean difference.
- Conclude if this variable is the same between the treatment and control groups.

Balance of the HH size

	Baseline 2011 (n = 832)					Endline 2014 (n = 959)				
	Control		Intervention		p^+	Control		Intervention		p^+
	n	mean ($\pm$ SD)	n	mean ($\pm$ SD)		n	mean ($\pm$ SD)	n	mean ($\pm$ SD)	
Household size	412	5.8 (2.3)	419	5.7 (2.1)	0.49	466	5.3 (1.9)	493	5.3 (2.0)	0.78
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- H0: Average household size is equal between Control and treatment;
 - $H_0 : \text{HH Size}_{\text{Control}} = \text{HH Size}_{\text{Treatment}}$
- H0: Average HH size is unequal
 - $H_1 : \text{HH Size}_{\text{Control}} \neq \text{HH Size}_{\text{Treatment}}$
- Probability of the type I error: 0.49 (p-value)
- Fail to reject H_0 .
- The difference between C and I is insignificant at the 5% level of significance.

Remember

The checks and balances table must indicate that treatment and control groups are similar on average.

- If they are not, simple means comparison will not reveal the causal effect of a treatment.

How to identify if the intervention helped?

Because Kuchenbecker et al. (2017) is an RCT, we can:

- Compare means of the outcome variable between two groups, by
- Performing statistical hypothesis testing (same as in checks and balances),
- But instead of looking for failing to reject the H_0 , look for a significant difference.

Intervention outcomes table

	Baseline 2011					Endline 2014						
	C	I		C	I	C	I		C	I	Intervention effect	
	(n = 413)	(n = 419)		(n = 413)	(n = 419)	(n = 466)	(n = 493)		(n = 466)	(n = 493)		
Food groups	prevalences unadjusted (%)		p^+	prevalences adjusted* (%)		prevalences unadjusted (%)		p^+	prevalences adjusted* (%)		(%)	p
Grains, roots & tubers	96.4	98.3	0.08	96.6	98.2	95.2	97.6	0.06	95.3	97.5	0.66	0.69
Grains	96.4	98.3	0.08	96.7	98.2	94.7	97.4	0.04	94.8	97.3	0.92	0.58
Roots & tubers	24.9	25.1	0.97	26.1	24.7	19.4	18.7	0.79	18.9	18.6	1.13	0.78
Legumes & nuts	57.1	67.5	<0.01	59.4	66.7	65.4	77.9	0.01	64.8	77.3	4.86	0.35
Groundnuts	35.8	46.3	<0.01	37.4	45.7	38.9	62.4	<0.01	38.7	61.9	14.96	<0.01
Other legumes & nuts	41.2	49.4	0.02	43.1	48.6	50.9	61.4	<0.01	50.5	60.9	4.87	0.34
Dairy products	12.9	13.4	0.83	13.5	12.5	15.6	20.9	0.03	16.2	20.6	5.47	0.15
Flesh foods	31.2	35.3	0.21	32.3	34.5	27.4	38.2	<0.01	27.0	38.4	9.20	0.09
Meat	15.5	16.7	0.64	16.0	15.9	12.8	18.9	0.01	13.0	19.1	6.18	0.11
Organ meat	2.7	2.8	0.80	2.8	2.3	1.3	2.6	0.14	1.2	2.6	1.96	0.16
Fish & seafood	19.9	23.2	0.25	20.6	22.9	18.5	24.1	0.04	18.2	24.1	3.57	0.43
Eggs	12.1	13.4	0.59	12.8	12.9	6.1	16.3	<0.01	6.1	16.2	9.92	<0.01
Vitamin A rich fruit & vegetables	69.0	75.7	0.03	71.1	75.6	71.6	77.5	0.04	70.2	77.3	2.62	0.53
Vitamin A rich roots & tubers	11.9	7.6	0.04	12.1	7.3	12.2	12.9	0.74	12.1	13.2	5.82	0.11
Vitamin A rich vegetables	65.9	73.0	0.03	68.0	72.9	67.8	73.3	0.07	66.4	73.1	1.77	0.67
Vitamin A rich fruit	10.4	16.2	0.01	10.8	16.1	6.7	16.9	<0.01	6.5	16.9	5.14	0.25
Other fruit & vegetables	72.4	78.3	<0.05	74.6	78.0	73.5	82.5	<0.01	72.5	82.0	6.16	0.11
IYCF indicators												
MDD	56.3	62.8	0.06	59.9	61.9	56.3	71.7	<0.01	55.5	71.1	12.70	0.01
MMF	80.0	89.2	<0.01	80.4	88.6	81.0	90.6	<0.01	81.6	90.3	0.43	0.91
MAD	47.9	57.7	<0.01	55.2	56.7	47.7	66.5	<0.01	47.3	65.8	11.86	0.02

Homework

- Watch: [How to Read Economics Research Papers: Randomized Controlled Trials \(RCTs\)](#)
 - See ([Carter, Greenberg, and Walker 2017](#)) that underpins the video.
- Practice:
 - Perform statistical hypothesis testing about the mean difference in Table 4 ([Kuchenbecker et al. 2017](#)).
 - Perform statistical hypothesis testing about the mean difference in Table 1 in ([Gertler 2004](#)).

t-test

Most commonly used statistical test for hypothesis testing about the estimated parameter (e.g., $\widehat{\text{mean}}$, but also other estimates).

The purpose:

Infer:

- whether or not the population parameter μ (mean, or else)
- is different from an arbitrary number (usually zero: $\mu_0 = 0$)
- based on our sample estimates of
 - the parameter ($\widehat{\text{mean}}$),
 - the variance (se or s), and
 - sample size n .

t-tests

In principal, there are:

- one-sample t-test: compares mean to a number
- two-sample t-test: compares means of two independent samples
 - for equal and unequal variances between samples

one-sample t-test: comparing mean to a number

The algorithm

1. **Step 1.** Hypothesis formulation;
2. **Step 2.** Calculating test statistic;
3. **Step 3.** Setting a decision rule for rejecting H_0 ;
4. **Step 4.** Making a conclusion;

Step 1. Hypothesis formulation

H_0 Null Hypothesis

The population mean equals to zero μ_0

- $H_0 : \mu = \mu_0$

H_1 Alternative Hypothesis

The population mean is not zero μ_0

- $H_1 : \mu \neq \mu_0$
- $H_1 : \mu < \mu_0$
- $H_1 : \mu > \mu_0$

Step 2. Calculating test statistic

t-statistics:

$$t = (\hat{\mu} - \mu_0) / \text{se} = \frac{\sqrt{n}(\hat{\mu} - \mu_0)}{s},$$

where:

- t - t-statistics;
- $\hat{\mu}$ - estimated mean and μ_0 is an arbitrary number;
- n - number of observation and s - sample standard deviation;

Sometimes, t-statistics is defined as:

$$t = (\hat{\mu} - \mu_0) / \text{se}(\hat{\mu}),$$

where:

- $\text{se}(\hat{\mu})$ is the standard error of the estimated coefficient $\hat{\mu}$
- $\text{se}(\hat{\mu}) = s / \sqrt{n}$;

Step 2. Example

Sample size: $n = 25$

Estimates mean: $\hat{\mu} = -2.2$

Sample SD: $s = 3.3$

- Compute the t-statistics for the mean difference from zero.
- $\mu_0 = 0?$
- $t = \frac{\sqrt{25}(-2.2-0)}{3.3}$
 $= 5 \cdot (-1/3)$
 $= -1.667$

- Compute the t-statistics for the mean difference from -5.5.
- $\mu_0 = -5.5?$
- $t = \frac{\sqrt{25}(-2.2+5.5)}{3.3}$
 $= 5 \cdot 1$
 $= 5$

Step 3. Setting a decision rule (1)

To reject H_0 , we compare t to a **critical value** $c_{df,CL}$, where:

- CL is the confidence level (opposite to the significance level: $1 - \alpha$)
- df - degrees of freedom

Step 3. Setting a decision rule (2)

Critical value, is the minimum values of the t-statistics at which we can reject H_0 at the given level of significance.

- Critical value depends on the alternative hypothesis:
- when $H_1 : \mu \neq \mu_0$ (two-sided t-test),
 - $CL = 1 - \alpha/2$ then $c_{df, 1-\alpha/2}$;
- $H_1 : \mu < \mu_0$ or $H_1 : \mu > \mu_0$ (one-sided t-test),
 - $CL = 1 - \alpha$ then $c_{df, 1-\alpha}$; ::

Step 3. Setting a decision rule (3)

Critical value also depends on:

- the number of degrees of freedom $df = n - k$:
 - n is the number of observation in the sample and
 - k is the number of parameters
 - $k = 1$ for one-sample t-test
 - $k = 2$ for two-sample t-test
- level of significance α ;
 - Conventionally $\alpha = 0.05$

Step 3. Setting a decision rule (4)

We reject $H_0 : \mu = \mu_0$ in favor of $H_1 : \mu \neq \mu_0$ **only if**:

- $|t| > c_{df, 1-\alpha/2}$ for two-sided t-test $H_1 : \mu \neq \mu_0$
- $|t| > c_{df, 1-\alpha}$ for one-sided t-test $H_1 : \mu > \mu_0$ or $H_1 : \mu < \mu_0$

Step 3. Setting a decision rule (4)

Finding the critical value is possible from the quantiles of the t-distribution.

- $df = n - k$;
- $CL = 1 - \alpha/2$ for $H_1 : \mu \neq \mu_0$
- $CL = 1 - \alpha$ for $H_1 : \mu > \mu_0$ or $H_1 : \mu < \mu_0$

df	0.9	0.95	0.975	0.99	0.995
1	3.0777	6.3138	12.7062	31.8205	63.6567
5	1.4759	2.0150	2.5706	3.3649	4.0321
10	1.3722	1.8125	2.2281	2.7638	3.1693
25	1.3163	1.7081	2.0595	2.4851	2.7874
40	1.3031	1.6839	2.0211	2.4233	2.7045
50	1.2987	1.6759	2.0086	2.4033	2.6778
75	1.2929	1.6654	1.9921	2.3771	2.6430
100	1.2901	1.6602	1.9840	2.3642	2.6259
150	1.2872	1.6551	1.9759	2.3515	2.6090
300	1.2844	1.6499	1.9679	2.3388	2.5923
1000	1.2824	1.6464	1.9623	2.3301	2.5808
10000	1.2816	1.6450	1.9602	2.3267	2.5763
1000000	1.2816	1.6449	1.9600	2.3264	2.5758

Step 3. Example (1)

Sample size: $n = 25$; $df = 25 - 1 = 24$

Is the population mean different from zero at a 1% level of significance?

- $H_1 : \mu \neq 0$
- $1 - \alpha/2 = 1 - 0.01/2 = 0.995$
- $c_{24, 1-0.01/2} = 2.7969$

df	0.95	0.975	0.99	0.995
1	6.3138	12.7062	31.8205	63.6567
5	2.0150	2.5706	3.3649	4.0321
10	1.8125	2.2281	2.7638	3.1693
20	1.7247	2.0860	2.5280	2.8453
23	1.7139	2.0687	2.4999	2.8073
24	1.7109	2.0639	2.4922	2.7969
25	1.7081	2.0595	2.4851	2.7874
26	1.7056	2.0555	2.4786	2.7787
27	1.7033	2.0518	2.4727	2.7707
28	1.7011	2.0484	2.4671	2.7633
29	1.6991	2.0452	2.4620	2.7564
30	1.6973	2.0423	2.4573	2.7500
50	1.6759	2.0086	2.4033	2.6778
150	1.6551	1.9759	2.3515	2.6090

Step 3. Example (2)

Sample size: $n = 25$; $df = 25 - 1 = 24$

Is the population mean different from zero at a 5% level of significance?

- $1 - \alpha/2 = 1 - 0.05/2 = 0.975$
- $c_{24, 1-0.05/2} = 2.0639$

df	0.95	0.975	0.99	0.995
1	6.3138	12.7062	31.8205	63.6567
5	2.0150	2.5706	3.3649	4.0321
10	1.8125	2.2281	2.7638	3.1693
20	1.7247	2.0860	2.5280	2.8453
23	1.7139	2.0687	2.4999	2.8073
24	1.7109	2.0639	2.4922	2.7969
25	1.7081	2.0595	2.4851	2.7874
26	1.7056	2.0555	2.4786	2.7787
27	1.7033	2.0518	2.4727	2.7707
28	1.7011	2.0484	2.4671	2.7633
29	1.6991	2.0452	2.4620	2.7564
30	1.6973	2.0423	2.4573	2.7500
50	1.6759	2.0086	2.4033	2.6778
150	1.6551	1.9759	2.3515	2.6090

Step 3. Example (3)

Sample size: $n = 25$; $df = 25 - 1 = 24$.

Is the population mean different from zero at 10% level of significance?

- $1 - \alpha/2 = 1 - 0.1/2 = 0.95$
- $c_{24, 1-0.1/2} = 1.7109$

df	0.95	0.975	0.99	0.995
1	6.3138	12.7062	31.8205	63.6567
5	2.0150	2.5706	3.3649	4.0321
10	1.8125	2.2281	2.7638	3.1693
20	1.7247	2.0860	2.5280	2.8453
23	1.7139	2.0687	2.4999	2.8073
24	1.7109	2.0639	2.4922	2.7969
25	1.7081	2.0595	2.4851	2.7874
26	1.7056	2.0555	2.4786	2.7787
27	1.7033	2.0518	2.4727	2.7707
28	1.7011	2.0484	2.4671	2.7633
29	1.6991	2.0452	2.4620	2.7564
30	1.6973	2.0423	2.4573	2.7500
50	1.6759	2.0086	2.4033	2.6778
150	1.6551	1.9759	2.3515	2.6090

Step 4. Make a conclusion

Concluding either or:

- We reject H_0
 - at least at the α level of significance.
 - The population mean is significantly different from μ_0 .
- We fail to reject (we accept) H_0 .
 - Available data do not provide sufficient evidence to reject H_0 at the α level of significance.
 - The population mean is statistically indifferent from μ_0 .

Step 4. Example (1)

Sample size: $n = 25$. Estimates mean: $\hat{\mu} = -2.2$. Sample SD: $s = 3.3$

Is the population mean different from zero ($\mu_0 = 0$) at 5% level of significance?

- $t = \frac{\sqrt{25}(-2.2-0)}{3.3} = -1.667$
- $c_{24, 1-0.05/2} = 2.0639$
- $|t| < c_{24, 1-0.05/2}$
- $|-1.667| < 2.0639$
- Conclusion:
 - Fail to reject (we accept) H_0 .
 - the population mean is statistically **indifferent** from $\mu_0 = 0$.

Step 4. Example (2)

Is the population mean different from -5.5 ($\mu_0 = -5.5$) at the 1% level of significance?

- $\mu_0 = -5.5$?
- $t = \frac{\sqrt{25}(-2.2+5.5)}{3.3} = 5$
- $c_{24, 1-0.05/2} = 2.7969$
- $|t| > c_{24, 1-0.01/2}$
- $|5| > 2.7969$
- Conclusion:
 - Reject H_0 .
 - the population mean is statistically different from $\mu_0 = -5.5$ at least at the 1% level of significance.

Homework (exam) examples (1)

Calculate the t-statistics and find corresponding critical values.

- Conclude whether the H_0 could be rejected in the following hypothesis testing problems.
- Compare your calculations to the solutions.

$\hat{\mu}$	s	μ_0	n	α	t	$c_{df,CL}$
4.0	1.0	3.0	10	0.01	3.162	3.169
2.0	1.2	0.8	8	0.05	2.828	2.306
3.5	3.1	2.8	150	0.01	2.766	2.609
8.0	4.0	16.0	21	0.10	-9.165	1.721
1.2	3.4	0.2	53	0.05	2.141	2.006

Critical values of the t-distribution.

df	0.95	0.975	0.99	0.995
1	6.3138	12.7062	31.8205	63.6567
5	2.0150	2.5706	3.3649	4.0321
6	1.9432	2.4469	3.1427	3.7074
7	1.8946	2.3646	2.9980	3.4995
8	1.8595	2.3060	2.8965	3.3554
9	1.8331	2.2622	2.8214	3.2498
10	1.8125	2.2281	2.7638	3.1693
20	1.7247	2.0860	2.5280	2.8453
25	1.7081	2.0595	2.4851	2.7874
30	1.6973	2.0423	2.4573	2.7500
50	1.6759	2.0086	2.4033	2.6778
80	1.6641	1.9901	2.3739	2.6387
100	1.6602	1.9840	2.3642	2.6259
150	1.6551	1.9759	2.3515	2.6090

Effects of a de-worming pill

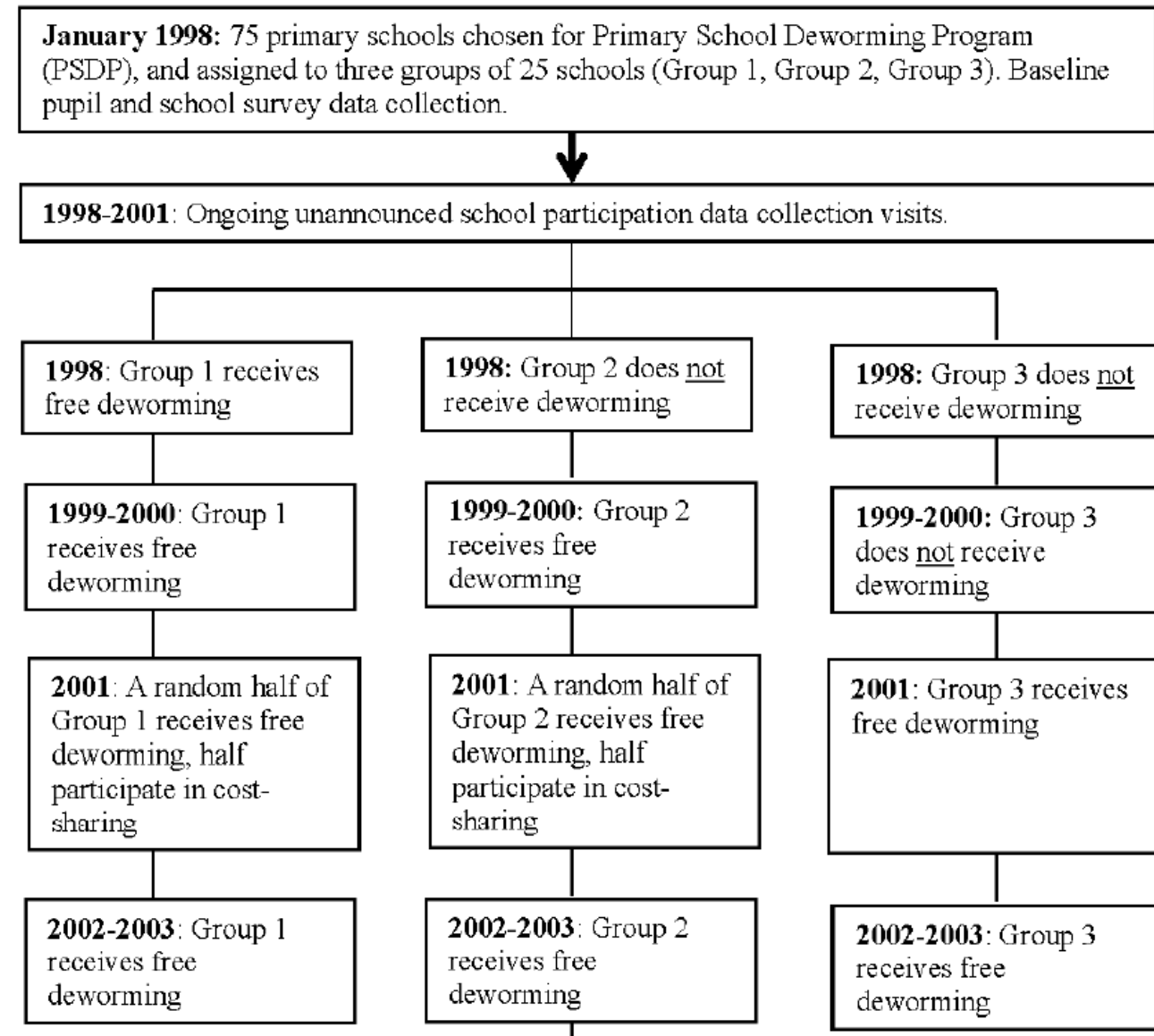
In 1998 an NGO initiated

- the Primary School Deworming Project [PSDP]
- in Busia District (Busia County), western Kenya that is fairly representative of rural Kenya
- in 75 randomly selected schools (32,000 pupils) with a helminths prevalence rate over 90%!

PSDP implementation timeline

Source: ([Hamory et al. 2021](#)), supplementary materials.

Fig. S1. Project Timeline of the Primary School Deworming Program (PSDP) and the Kenya Life Panel Survey (KLPS)



Checks and balances between groups

Source: (Miguel and
Kremer 2004) table 1.

- See the full table and interpret it.
- Is there any difference between treatment and control groups?

TABLE I					
1998 AVERAGE PUPIL AND SCHOOL CHARACTERISTICS, PRE-TREATMENT ^a					
	Group 1 (25 schools)	Group 2 (25 schools)	Group 3 (25 schools)	Group 1 – Group 3	Group 2 – Group 3
<i>Panel A: Pre-school to Grade 8</i>					
Male	0.53	0.51	0.52	0.01 (0.02)	–0.01 (0.02)
Proportion girls <13 years, and all boys	0.89	0.89	0.88	0.00 (0.01)	0.01 (0.01)
Grade progression (= Grade – (Age – 6))	–2.1	–1.9	–2.1	–0.0 (0.1)	0.1 (0.1)
Year of birth	1986.2	1986.5	1985.8	0.4** (0.2)	0.8*** (0.2)
<i>Panel B: Grades 3 to 8</i>					
Attendance recorded in school registers (during the four weeks prior to the pupil survey)	0.973	0.963	0.969	0.003 (0.004)	–0.006 (0.004)
Access to latrine at home	0.82	0.81	0.82	0.00 (0.03)	–0.01 (0.03)
Have livestock (cows, goats, pigs, sheep) at home	0.66	0.67	0.66	–0.00 (0.03)	0.01 (0.03)
Weight-for-age Z-score (low scores denote undernutrition)	–1.39	–1.40	–1.44	0.05 (0.05)	0.04 (0.05)
Blood in stool (self-reported)	0.26	0.22	0.19	0.07** (0.03)	0.03 (0.03)
Sick often (self-reported)	0.10	0.10	0.08	0.02** (0.01)	0.02** (0.01)
Malaria/fever in past week (self-reported)	0.37	0.38	0.40	–0.03 (0.03)	–0.02 (0.03)

Immediate effect on health

Source: (Miguel and Kremer 2004) Table 5.

- Check the full table and interpret it.
- What effects were and were not caused by deworming treatment?

TABLE V			
JANUARY TO MARCH 1999, HEALTH AND HEALTH BEHAVIOR DIFFERENCES BETWEEN GROUP 1 (1998 TREATMENT) AND GROUP 2 (1998 COMPARISON) SCHOOLS ^a			
	Group 1	Group 2	Group 1 – Group 2
<i>Panel A: Helminth Infection Rates</i>			
Any moderate-heavy infection, January–March 1998	0.38	–	–
Any moderate-heavy infection, 1999	0.27	0.52	–0.25*** (0.06)
Hookworm moderate-heavy infection, 1999	0.06	0.22	–0.16*** (0.03)
Roundworm moderate-heavy infection, 1999	0.09	0.24	–0.15*** (0.04)
Schistosomiasis moderate-heavy infection, 1999	0.08	0.18	–0.10* (0.06)
Whipworm moderate-heavy infection, 1999	0.13	0.17	–0.04 (0.05)
<i>Panel B: Other Nutritional and Health Outcomes</i>			
Sick in past week (self-reported), 1999	0.41	0.45	–0.04** (0.02)
Sick often (self-reported), 1999	0.12	0.15	–0.03** (0.01)
Height-for-age Z-score, 1999 (low scores denote undernutrition)	–1.13	–1.22	0.09* (0.05)
Weight-for-age Z-score, 1999 (low scores denote undernutrition)	–1.25	–1.25	–0.00 (0.04)
Hemoglobin concentration (g/L), 1999	124.8	123.2	1.6 (1.4)
Proportion anemic (Hb < 100g/L), 1999	0.02	0.04	–0.02** (0.01)

Immediate effect on school participation

Source: (Miguel and Kremer 2004) Table 8.

- Check the full table and interpret it.

TABLE VIII
SCHOOL PARTICIPATION, SCHOOL-LEVEL DATA^a

	Group 1 (25 schools)	Group 2 (25 schools)	Group 3 (25 schools)		
<i>Panel A:</i>					
<i>First year post-treatment (May 1998 to March 1999)</i>	<i>1st Year Treatment</i>	<i>Comparison</i>	<i>Comparison</i>	<i>Group 1 – (Groups 2 & 3)</i>	<i>Group 2 – Group 3</i>
Girls <13 years, and all boys	0.841	0.731	0.767	0.093*** (0.031)	–0.037 (0.036)
Girls ≥13 years	0.864	0.803	0.811	0.057** (0.029)	–0.008 (0.034)
Preschool, Grade 1, Grade 2 in early 1998	0.795	0.688	0.703	0.100*** (0.037)	–0.018 (0.043)
Grade 3, Grade 4, Grade 5 in early 1998	0.880	0.789	0.831	0.070*** (0.024)	–0.043 (0.029)
Grade 6, Grade 7, Grade 8 in early 1998	0.934	0.858	0.892	0.059*** (0.021)	–0.034 (0.026)
Recorded as “dropped out” in early 1998	0.064	0.050	0.030	0.022 (0.018)	0.020 (0.017)
Females ^b	0.855	0.771	0.789	0.076*** (0.027)	–0.018 (0.032)
Males	0.844	0.736	0.780	0.088*** (0.031)	–0.044 (0.037)
<i>Panel B:</i>					

Long-term economic effects (1)

Source: (Hamory et al. 2021) Table 1.

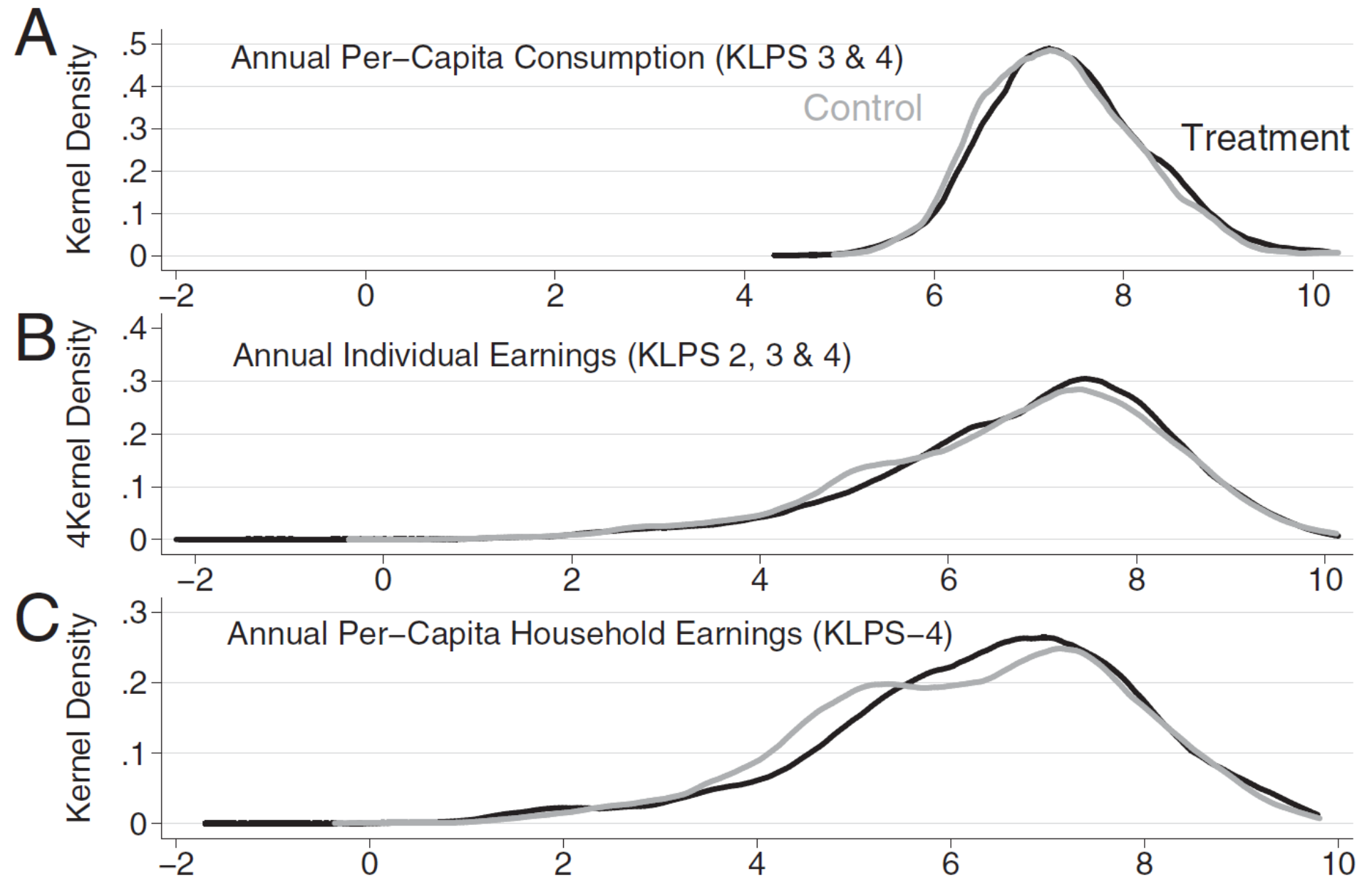
- Check the full table and interpret it.

	(1) Full sample	(2) Female	(3) Male	(4) Older	(5) Younger
A: Annual per capita consumption (KLPS-3 and KLPS-4)					
Treatment (λ_1)	305* (159)	89 (134)	513* (304)	886*** (223)	−179 (185)
Control mean	2,156	1,715	2,594	1,908	2,381
Treatment effect (%)	14.15	5.21	19.76	46.44	−7.52
Treatment <i>P</i> value	0.058	0.505	0.096	0.000	0.337
FDR <i>q</i> value	0.132	0.630	0.623	0.001	0.290
Number observations	4,794	2,473	2,321	2,402	2,341
B: Annual individual earnings (KLPS-2, KLPS-3, and KLPS-4)					
Treatment (λ_1)	80 (76)	41 (62)	118 (133)	258** (108)	−75 (100)
Control mean	1,218	674	1,728	1,177	1,242
Treatment effect (%)	6.53	6.02	6.84	21.93	−6.07
Treatment <i>P</i> value	0.297	0.515	0.376	0.019	0.451
FDR <i>q</i> value	0.175	0.630	0.630	0.030	0.292
Number of observations	13,624	6,826	6,798	6,791	6,780
C: Annual per capita household earnings (KLPS-4)					
Treatment (λ_1)	239* (129)	36 (107)	439* (252)	565** (232)	−22 (171)
Control mean	1,296	973	1,623	1,082	1,501
Treatment effect (%)	18.44	3.68	27.06	52.17	−1.48
Treatment <i>P</i> value	0.069	0.738	0.086	0.017	0.897
Number of observations	4,074	2,099	1,975	2,039	1,982

Long-term economic effects (2)

Source:
([Hamory et al. 2021](#))
Figure 1.

- the x-axis represents the log of annual earnings per capita.



Long-term economic effects (3)

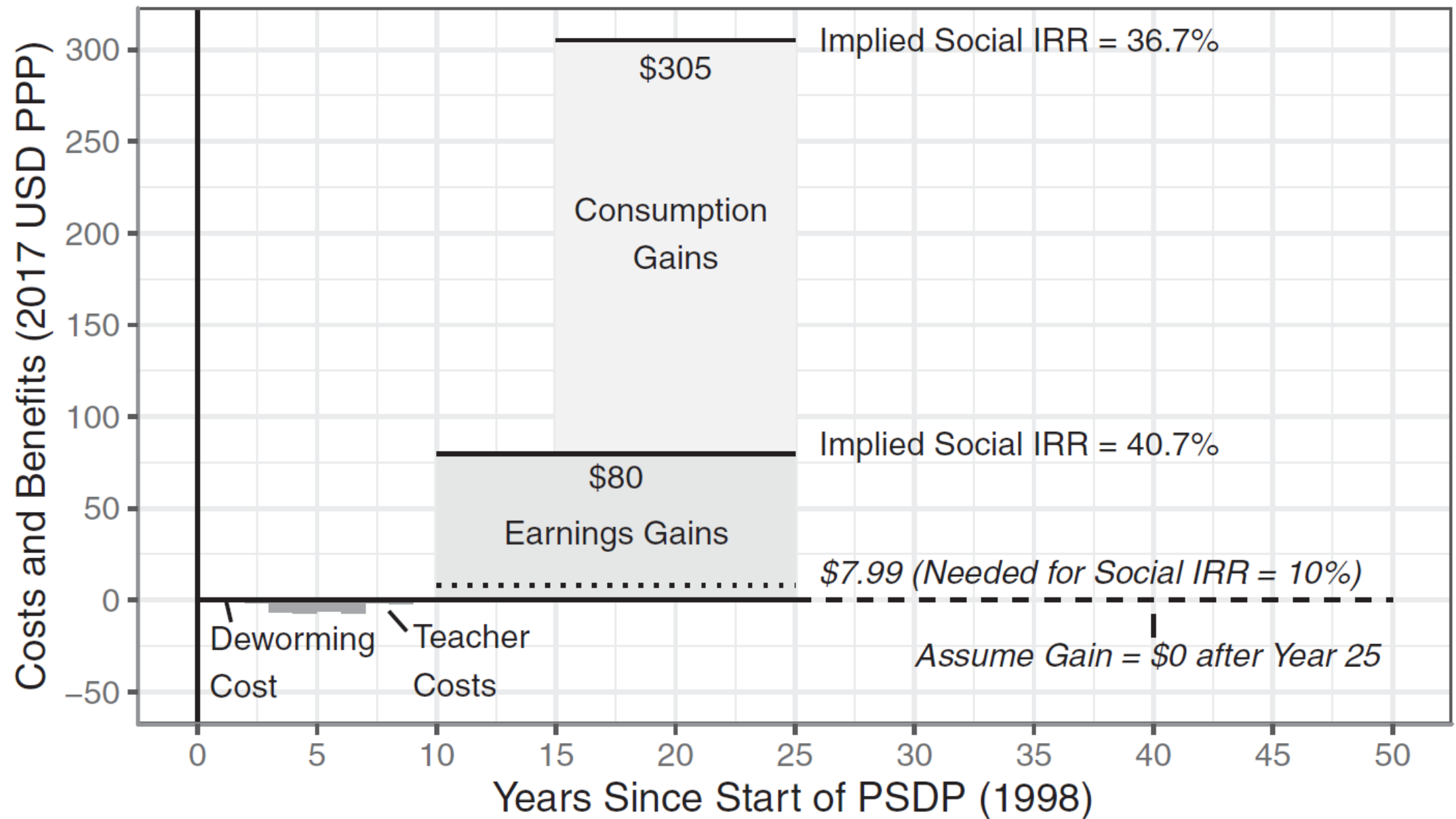
Source: ([Hamory et al. 2021](#)) Table 2.

Table 2. The 10- to 20-y deworming treatment effects on earnings, labor supply, occupation, and sectoral choice, KLPS-2, KLPS-3, and KLPS-4

	Treatment (λ_1)			Full sample	
	(1) Full sample	(2) Male	(3) Older	(4) Control mean	(5) Number of obs.
A: Earnings and wealth					
Log annual individual earnings	0.09 (0.06)	0.06 (0.07)	0.19** (0.08)	6.73	7,698
Wage earnings (annual)	81 (68)	138 (110)	162* (89)	887	13,628
Self-employment profit (annual)	41* (24)	51 (48)	70* (39)	212	13,638
Individual farming profit (annual)	−0 (2)	1 (3)	−3 (3)	9	13,707
Nonzero earnings	0.02* (0.01)	0.04** (0.02)	0.02 (0.02)	0.59	13,794
Hourly earnings	0.14* (0.08)	0.22 (0.15)	0.32* (0.16)	1.07	6,096
Per capita household wealth (KLPS-4)	69 (50)	102 (97)	253*** (89)	522	4,085
B: Labor supply, occupation, and sectoral choice					
Urban residence	0.04** (0.02)	0.06** (0.03)	0.03 (0.03)	0.45	13,793
Total hours worked (last 7 d)	1.04 (0.66)	2.20** (0.92)	1.79** (0.91)	24.19	13,807
Hours worked—agriculture (last 7 d)	−0.87** (0.43)	−0.57 (0.62)	−0.46 (0.56)	3.99	13,807
Hours worked—nonagriculture (last 7 d)	1.91*** (0.65)	2.77*** (0.94)	2.24** (1.08)	20.20	13,807
Employed—agriculture/fishing	−0.003 (0.008)	−0.001 (0.013)	0.004 (0.012)	0.043	13,768
Employed—services/wholesale/retail	0.002 (0.014)	0.012 (0.020)	−0.002 (0.019)	0.230	13,761
Employed—construction/trade contractor	0.004 (0.007)	0.011 (0.014)	−0.007 (0.009)	0.033	13,760
Employed—manufacturing	−0.001 (0.004)	0.002 (0.007)	0.002 (0.006)	0.026	13,760

Costs and benefits of deworming

Source:
([Hamory
et al.
2021](#))
Figure
2



References

Takeaways checklist

Homework

Make the topic-specific homework.

Answer sample exam questions.

Same readings:

- ([Kuchenbecker et al. 2017](#)) **before the week 4;**
- ([Gertler 2004](#)) another example of the statistics application;
- ([Miguel and Kremer 2004](#)) Deworming effect on children health;
- ([Hamory et al. 2021](#)) Long-term economic effect of de-worming;
- ([Carter, Greenberg, and Walker 2017](#)) Effect of electronics on students performance.

Key concepts covered in the lecture

RCT:

- Causal meaning of the averages;
- Random sample variation;
- Need of the statistical hypothesis testing;

Hypothesis testing:

- Type I and Type II errors;
- Significance level;
- Null and Alternative Hypotheses;
- What is the probability value (p-value)?
- What do false-negative/false-positive mean?

T-test:

- One-sided/Two-sided.
- Critical values of the t-distributions
- t statistics
- Mechanics of the test statistics calculation.

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